

Conditions

```
"""  
  
if conditions:  
    statement.....  
  
elif conditions:  
    statement.....  
  
else:  
    statement.....  
  
"""
```

if - else

```
In [2]:  ▶ a = 6  
  
        if a == 8:  
            print(a, 'is equal to 8')  
        else:  
            print(a, 'is not equal to 8')
```

6 is not equal to 8

```
In [20]: ▶ # user input  
  
a = int(input('Enter any number: '))  
a
```

Enter any number: 4

Out[20]: 4

```
In [19]: ► a = eval(input('Enter any number: '))  
a
```

Enter any number: [4, 8, 6, 7]

Out[19]: [4, 8, 6, 7]

```
In [ ]: ►
```

```
In [25]: ► age = int(input('Enter any number: '))  
  
print()  
  
if age < 18:  
    print('Person is not eligible for vote.')  
else:  
    print('Person is eligible for vote.')
```

Enter any number: 21

Person is eligible for vote.

```
In [ ]: ►
```

if - elif - else

```
In [29]: x = int(input('Enter any number: '))

print()

if x < 0:
    print('X is Negative')

elif x == 0:
    print('X is zero')

else:
    print('X is Positive')
```

Enter any number: 0

X is zero

```
In [ ]:
```

```
In [30]: x = int(input('Enter any number: '))

print()

if x < 0:
    print('X is Negative')

elif x == 0:
    print('X is zero')

elif x > 0 and x <= 20:
    print('X is between 0 and 20')

else:
    print('X is greater than 20')
```

Enter any number: 8

X is between 0 and 20

In []: ▶

Nested if-else

```
In [34]: ▶ x = int(input('Enter any number: '))

if x >= 10:

    if x <= 20:
        print('X is between 10 and 20')
    else:
        print('X is greater than 20')

else:
    print('X is less than 10')
```

Enter any number: 56
X is greater than 20

In []: ▶

```
In [46]: x = int(input('Enter any number: '))

if x >= 10:

    if x <= 20:
        print('X is between 10 and 20')

    elif x > 20:

        if x <= 50:
            print('X is between 20 and 50')
        elif x > 50 and x <= 100:
            print('X is between 50 and 100')
        else:
            print('X is greater than 100')

else:
    print('X is less than 10')
```

Enter any number: 205
X is greater than 100

In []:

In []:

Iterative Statements - Loops

```
In [ ]: # while loop
        # for loop
```

In []:

while loop

```
In [2]: ▶ a = 0  
  
while a <= 10:  
    print(a)  
    a += 1
```

```
0  
1  
2  
3  
4  
5  
6  
7  
8  
9  
10
```

```
In [6]: ▶ a = 0

while a <= 10:
    print('Loop is still running.')
    # print(a)
    a += 1

print()
print('Loop has Terminated')
```

Loop is still running.
Loop is still running.
Loop is still running.
Loop is still running.
Loop is still running.
Loop is still running.
Loop is still running.
Loop is still running.
Loop is still running.
Loop is still running.
Loop is still running.

Loop has Terminated

```
In [ ]: ▶
```

For Loop

```
In [13]: ▶ lst = [2,4,6,3,2,1,5,5,32,2,2,5,6,7,45,6,76,4,5,7,8]  
lst
```

```
for item in lst:  
    print(item)
```

```
2  
4  
6  
3  
2  
1  
5  
5  
32  
2  
2  
5  
6  
7  
45  
6  
76  
4  
5  
7  
8
```

```
In [12]: ▶ list(iter(lst))
```

```
Out[12]: [2, 4, 6, 3, 2, 1, 5, 5, 32, 2, 2, 5, 6, 7, 45, 6, 76, 4, 5, 7, 8]
```

```
In [ ]: ▶
```



```
In [14]: ► fruits = ['Apple', 'Banana', 'Mango', 'Kiwi', 'Orange', 'Grapes', 'Strawberry']

for fruit in fruits:
    print(fruit)
```

Apple
Banana
Mango
Kiwi
Orange
Grapes
Strawberry

```
In [20]: ► fruits = ['Apple', 'Banana', 'Mango', 'Kiwi', 'Orange', 'Grapes', 'Strawberry']

for fruit in fruits:
    if fruit == 'Orange':
        print(fruit, 'present')
    else:
        print('Not Present')
```

Not Present
Not Present
Not Present
Not Present
Orange present
Not Present
Not Present

```
In [ ]: ►
```

Statements

Break Statement

```
In [27]: ► fruits = ['Apple', 'Banana', 'Mango', 'Kiwi', 'Orange', 'Grapes', 'Strawberry']

for fruit in fruits:
    if fruit == 'Banana':
        print(fruit, 'present')
        break
    else:
        print('Not Present')
```

Not Present
Banana present

In []: ►

```
In [34]: ► fruits = ['Apple', 'Banana', 'Mango', 'Kiwi', 'Orange', 'Grapes', 'Strawberry']

L = []

for fruit in fruits:
    if fruit == 'Banana':
        L.append(fruit)
        break

L
```

Out[34]: ['Banana']

```
In [36]: ► fruits = ['Apple', 'Banana', 'Mango', 'Kiwi', 'Orange', 'Grapes', 'Strawberry']

L = []

for fruit in fruits:
    if fruit == 'Orange':
        L.append(fruit)
        break
    else:
        print(fruit)

L
```

Apple
Banana
Mango
Kiwi

Out[36]: ['Orange']

In []: ►

Continue Statement

In [42]: `fruits = ['Apple', 'Banana', 'Mango', 'Kiwi', 'Orange', 'Grapes', 'Strawberry']`

`L = []`

```
for fruit in fruits:
    if fruit == 'Kiwi':
        continue
    print(fruit)
```

Apple
Banana
Mango
Orange
Grapes
Strawberry

In [45]: `fruits = ['Apple', 'Banana', 'Mango', 'Kiwi', 'Orange', 'Grapes', 'Kiwi', 'Strawberry', 'Kiwi']`

`L = []`

```
for fruit in fruits:
    if fruit == 'Kiwi':
        continue
    L.append(fruit)
```

`L`

Out[45]: `['Apple', 'Banana', 'Mango', 'Orange', 'Grapes', 'Strawberry']`

```
In [48]: ► fruits = ['Apple', 'Banana', 'Mango', 'Kiwi', 'Orange', 'Grapes', 'Kiwi', 'Strawberry', 'Kiwi']

L = []

for fruit in fruits:
    if fruit != 'Kiwi':
        continue
    L.append(fruit)

L
```

```
Out[48]: ['Kiwi', 'Kiwi', 'Kiwi']
```

```
In [ ]: ►
```

Pass Statement

```
In [50]: ► fruits = ['Apple', 'Banana', 'Mango', 'Kiwi', 'Orange', 'Grapes', 'Kiwi', 'Strawberry', 'Kiwi']

L = []

for fruit in fruits:
    if fruit != 'Kiwi':
        pass
    L.append(fruit)

L
```

```
Out[50]: ['Apple',
          'Banana',
          'Mango',
          'Kiwi',
          'Orange',
          'Grapes',
          'Kiwi',
          'Strawberry',
          'Kiwi']
```

```
In [51]: ► fruits = ['Apple', 'Banana', 'Mango', 'Kiwi', 'Orange', 'Grapes', 'Kiwi', 'Strawberry', 'Kiwi']

L = []

for fruit in fruits:
    if fruit == 'Kiwi':
        pass
    L.append(fruit)

L
```

```
Out[51]: ['Apple',
          'Banana',
          'Mango',
          'Kiwi',
          'Orange',
          'Grapes',
          'Kiwi',
          'Strawberry',
          'Kiwi']
```

```
In [ ]: ►
```

```
In [55]: ► def my_func():
           pass
```

```
In [ ]: ►
```

Exception Handling

In [63]: `x = int(input('Enter any number: '))`

```
div = 100/x
div
```

Enter any number: 0

```
-----
ZeroDivisionError                                Traceback (most recent call last)
Cell In[63], line 3
      1 x = int(input('Enter any number: '))
----> 3 div = 100/x
      4 div
```

ZeroDivisionError: division by zero

In []:

In [70]: `try:`

```
x = int(input('Enter any number: '))
div = 100/x
print(div)
```

```
except:
    print('Error')
```

Enter any number: 0
Error

In []:

In [76]: ▶ try:

```
x = int(input('Enter any number: '))  
div = 100/x  
print(div)
```

```
except:  
    pass
```

Enter any number: 6
16.666666666666668

In []: ▶

In [78]: ▶ try:

```
x = int(input('Enter any number: '))  
div = 100/x  
print(div)
```

```
except Exception as e:  
    print(e)
```

Enter any number: 0
division by zero

In []: ▶

In [79]: ▶ lst = [2,4,32,34,5,5,7]
lst

Out[79]: [2, 4, 32, 34, 5, 5, 7]

In [80]: `lst[2]`

Out[80]: 32

In [83]: `lst[8]`

```
-----  
IndexError                                Traceback (most recent call last)  
Cell In[83], line 1  
----> 1 lst[8]  
  
IndexError: list index out of range
```

In []:

In [86]: `try:`
 `lst = [2,4,32,34,5,5,7]`
 `print(lst[8])`

 `except Exception as e:`
 `print(e)`

list index out of range

In []:

File Handling

```
In [103]: ► open('abc.txt', mode='r')
```

```
-----  
FileNotFoundError                                Traceback (most recent call last)  
Cell In[103], line 1  
----> 1 open('abc.txt', mode='r')  
  
File ~\anaconda3\Lib\site-packages\IPython\core\interactiveshell.py:284, in _modified_open(file, *args, **kwargs)  
    277 if file in {0, 1, 2}:  
    278     raise ValueError(  
    279         f"IPython won't let you open fd={file} by default "  
    280         "as it is likely to crash IPython. If you know what you are doing, "  
    281         "you can use builtins' open."  
    282     )  
--> 284 return io_open(file, *args, **kwargs)  
  
FileNotFoundError: [Errno 2] No such file or directory: 'abc.txt'
```

```
In [104]: file = open('abc.txt', mode='r')
file.close()
```

```
-----
FileNotFoundError                                Traceback (most recent call last)
Cell In[104], line 1
----> 1 file = open('abc.txt', mode='r')
      2 file.close()

File ~\anaconda3\Lib\site-packages\IPython\core\interactiveshell.py:284, in _modified_open(file, *args,
**kwargs)
    277 if file in {0, 1, 2}:
    278     raise ValueError(
    279         f"IPython won't let you open fd={file} by default "
    280         "as it is likely to crash IPython. If you know what you are doing, "
    281         "you can use builtins' open."
    282     )
--> 284 return io_open(file, *args, **kwargs)

FileNotFoundError: [Errno 2] No such file or directory: 'abc.txt'
```

```
In [105]: file = open('sourav.txt', mode='r')
file.close()
```

```
In [106]: file = open('sourav.txt', mode='r')
file.readline()
```

```
Out[106]: 'I am a Data Scientist.\n'
```

```
In [107]: file = open('sourav.txt', mode='r')
file.readlines()
```

```
Out[107]: ['I am a Data Scientist.\n',
'I am working on AI and Machine Learning.\n',
'Training Basket.']
```

```
In [108]: file = open('sourav.txt', mode='r')
          file.read()
          file.close()
```

```
In [109]: file.read()
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[109], line 1
----> 1 file.read()

ValueError: I/O operation on closed file.
```

```
In [110]: file = open('sourav.txt', mode='r')
          file.read()
```

```
Out[110]: 'I am a Data Scientist.\nI am working on AI and Machine Learning.\nTraining Basket.'
```

```
In [ ]: 
```

```
In [111]: file = open('D:\my_file.txt', mode='r')
          file.read()
          file.close()
```

```
In [ ]: 
```

```
In [112]: file = open('D:\\PPTs\\file_2.txt', mode='r')
          file.read()
```

```
Out[112]: 'This is another file.'
```

In []: ▶

```
In [113]: ▶ with open('D:\\PPTs\\file_2.txt', mode='r') as filename:
            content = filename.read()

            content
```

Out[113]: 'This is another file.'

```
In [115]: ▶ with open('sourav.txt', mode='r') as filename:
            content = filename.readlines()

            content
```

Out[115]: ['I am a Data Scientist.\n',
 'I am working on AI and Machine Learning.\n',
 'Training Basket.']

```
In [116]: ▶ with open('sourav.txt', mode='r') as filename:
            for line in filename:
                print(line)
```

I am a Data Scientist.

I am working on AI and Machine Learning.

Training Basket.

In []: ▶

Write a new file

```
In [117]: ► with open('deepak.txt', mode='w') as deepakfile:
            deepakfile.write('My name is Deepak and I am a data analyst.')
```

```
In [119]: ► with open('deepak.txt', mode='r') as deepakfile:
            data = deepakfile.read()

            data
```

Out[119]: 'My name is Deepak and I am a data analyst.'

```
In [120]: ► with open('deepak.txt', mode='w') as deepakfile:
            deepakfile.write('I have done with my assignment.')
```

```
In [121]: ► with open('deepak.txt', mode='r') as deepakfile:
            data = deepakfile.read()

            data
```

Out[121]: 'I have done with my assignment.'

```
In [ ]: ►
```

Append new data

```
In [122]: ► with open('deepak.txt', mode='r') as deepakfile:
            data = deepakfile.read()

            data
```

Out[122]: 'I have done with my assignment.'

```
In [123]: ► with open('deepak.txt', mode='a') as deepakfile:
           deepakfile.write('My name is Deepak and I am a data analyst.')
```

```
In [124]: ► with open('deepak.txt', mode='r') as deepakfile:
           data = deepakfile.read()

           data
```

```
Out[124]: 'I have done with my assignment.My name is Deepak and I am a data analyst.'
```

```
In [ ]: ►
```

Delete data in a text file

```
In [126]: ► with open('deepak.txt', 'w') as deepakfile:
           deepakfile.truncate()
```

```
In [127]: ► with open('deepak.txt', mode='r') as deepakfile:
           data = deepakfile.read()

           data
```

```
Out[127]: ''
```

OS module

```
In [128]: ► import os
```

```
In [129]: ► os.path.exists('rahul.txt')
```

```
Out[129]: False
```

```
In [130]: ► os.path.exists('sourav.txt')
```

```
Out[130]: True
```

```
In [ ]: ►
```

```
In [132]: ► if os.path.exists('rahul.txt'):  
            print('Exist')  
        else:  
            print('Not Exist')
```

```
Not Exist
```



```
In [152]: ▶ if os.path.exists('deepak.txt'):
           print('Exist')
           os.remove("deepak.txt")

           else:
           print('Not Exist')
```

Exist

PermissionError

Traceback (most recent call last)

Cell In[152], line 3

```
1 if os.path.exists('deepak.txt'):
2     print('Exist')
----> 3     os.remove("deepak.txt")
5 else:
6     print('Not Exist')
```

PermissionError: [WinError 32] The process cannot access the file because it is being used by another process: 'deepak.txt'

```
In [ ]: ▶
```

```
In [ ]: ▶
```

```
In [134]: ▶ with open('sourav.txt', 'r') as file:
           print(file.read())
```

I am a Data Scientist.
I am working on AI and Machine Learning.
Training Basket.

```
In [135]: >> with open('abc.txt', 'r') as file:
           print(file.read())
```

```
-----
FileNotFoundError                                Traceback (most recent call last)
Cell In[135], line 1
----> 1 with open('abc.txt', 'r') as file:
      2     print(file.read())

File ~\anaconda3\Lib\site-packages\IPython\core\interactiveshell.py:284, in _modified_open(file, *args,
**kwargs)
    277 if file in {0, 1, 2}:
    278     raise ValueError(
    279         f"IPython won't let you open fd={file} by default "
    280         "as it is likely to crash IPython. If you know what you are doing, "
    281         "you can use builtins' open."
    282     )
--> 284 return io_open(file, *args, **kwargs)

FileNotFoundError: [Errno 2] No such file or directory: 'abc.txt'
```

```
In [138]: >> try:
           with open('abc.txt', 'r') as file:
               print(file.read())

           except Exception as e:
               print(e)
```

```
[Errno 2] No such file or directory: 'abc.txt'
```

```
In [ ]: >>
```

```
In [ ]: >>
```

In []: 

In []: 

In []: 

In []: 

In []: 

In []: 

In []: 